Misurazione Implicita in Psicologia:
L'analisi classica dei dati IAT

Ottavia M. Epifania
University of Trento
ottavia.epifania@unitn.it

Master di II livello in Psicologia quantitativa

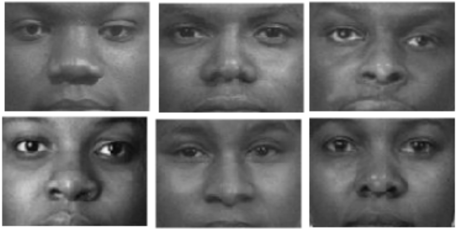
11 Settembre 2026

Implicit Association Test

12 stimoli target

White people faces

Black people faces



16 stimoli attributo

Attributi positivi

Good, laughter, pleasure, glory, peace, happy, joy, love

Attributi negativi

Evil, bad, horrible, terrible, nasty, pain, failure, hate

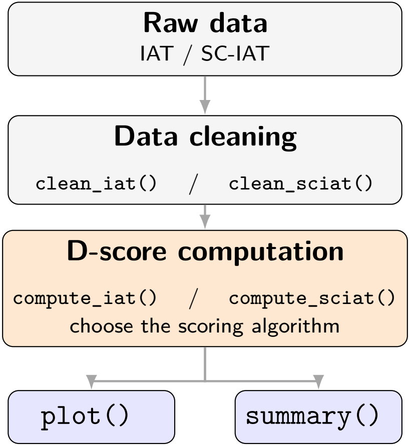
Person-level scores

$$s_p = \frac{\bar{X}_{p,\text{comp}} - \bar{X}_{p,\text{inc}}}{sd_{\text{pooled}}}$$

D score	Risposte errate	Risposte troppo veloci
D1	Built-in correction	No
D2	Built-in correction	Eliminare trials < 400 *ms*
D3	Mean (correct responses) + 2*sd*	No
D4	Mean (correct responses) + 600 *ms*	No
D5	Mean (correct responses) + 2*sd*	Eliminare trials < 400 *ms*
D6	Mean (correct responses) + 600 *ms*	Eliminare trials < 400 *ms*

Workflow

```r
install.packages("implicitMeasures")
library(implicitMeasures)
```


Raw data
IAT / SC-IAT
Data cleaning
clean_iat() / clean_sciat()
D-score computation
compute_iat() / compute_sciat()
choose the scoring algorithm
plot()
summary()

```r
data("raw_data") # load data
str(raw_data)
```

```
'data.frame':   84726 obs. of  6 variables:
 $ Participant: int  4 4 4 4 4 4 4 4 4 4 ...
 $ latency    : int  2592 628 808 783 2059 1114 608 663 771 676 ...
 $ correct    : int  1 1 1 1 1 1 1 1 1 1 ...
 $ trialcode  : Factor w/ 32 levels "age","alert",..: 31 5 3 20 3 20
 $ blockcode  : Factor w/ 13 levels "demo","practice.iat.Milkbad",..
 $ response   : Factor w/ 46 levels "","0","1","19",..: 43 43 43 43
```

```r
iat_cleandata <- clean_iat(raw_data, sbj_id = "Participant",
                          block_id = "blockcode",
                          mapA_practice = "practice.iat.Milkbad",
                          mapA_test = "test.iat.Milkbad",
                          mapB_practice = "practice.iat.Milkgood",
                          mapB_test = "test.iat.Milkgood",
                          latency_id = "latency",
                          accuracy_id = "correct",
                          trial_id = "trialcode",
                          trial_eliminate = c("reminder", "reminder1"),
                          demo_id = "blockcode",
                          trial_demo = "demo")
```

```r
str(iat_cleandata)
```

```
List of 3
 $ data_keep     :Classes 'iat_clean' and 'data.frame': 19440 obs. of  8 variables:
  ..$ participant    : int [1:19440] 4 4 4 4 4 4 4 4 4 4 ...
  ..$ latency        : int [1:19440] 1282 1299 1435 1089 967 648 967 615 729 642 ...
  ..$ correct        : int [1:19440] 1 1 1 1 1 1 1 1 1 1 ...
  ..$ block_original: chr [1:19440] "practice.iat.Milkbad" "practice.iat.Milkbad" '
  ..$ condition      : chr [1:19440] "MappingA" "MappingA" "MappingA" "MappingA" ...
  ..$ block_pool     : chr [1:19440] "practice" "practice" "practice" "practice" ...
  ..$ block          : chr [1:19440] "practice_MappingA" "practice_MappingA" "practi
  ..$ trial_id       : Factor w/ 32 levels "age","alert",..: 20 6 20 6 20 6 20 6 20
 $ data_eliminate:'data.frame': 64638 obs. of  6 variables:
  ..$ Participant: int [1:64638] 4 4 4 4 4 4 4 4 4 4 ...
  ..$ latency    : int [1:64638] 2592 628 808 783 2059 1114 608 663 771 676 ...
  ..$ correct    : int [1:64638] 1 1 1 1 1 1 1 1 1 1 ...
  ..$ trialcode  : Factor w/ 32 levels "age","alert",..: 31 5 3 20 3 20 5 3 20 3 ..
  ..$ blockcode  : Factor w/ 13 levels "demo","practice.iat.Milkbad",..: 4 4 4 4 4
  ..$ response   : Factor w/ 46 levels "","0","1","19",..: 43 43 43 43 43 43 43 43
 $ demo          :'data.frame': 3726 obs. of  6 variables:
  ..$ participant: int [1:3726] 4 4 4 4 4 4 4 4 4 4 ...
  ..$ latency    : int [1:3726] 53047 53047 53047 53047 21554 21554 11266 11266 112
  ..$ correct    : int [1:3726] 1 1 1 1 1 1 1 1 1 1
```

```
dscore = compute_iat(iat_cleandata, Dscore = "d1")
summary(dscore)
```

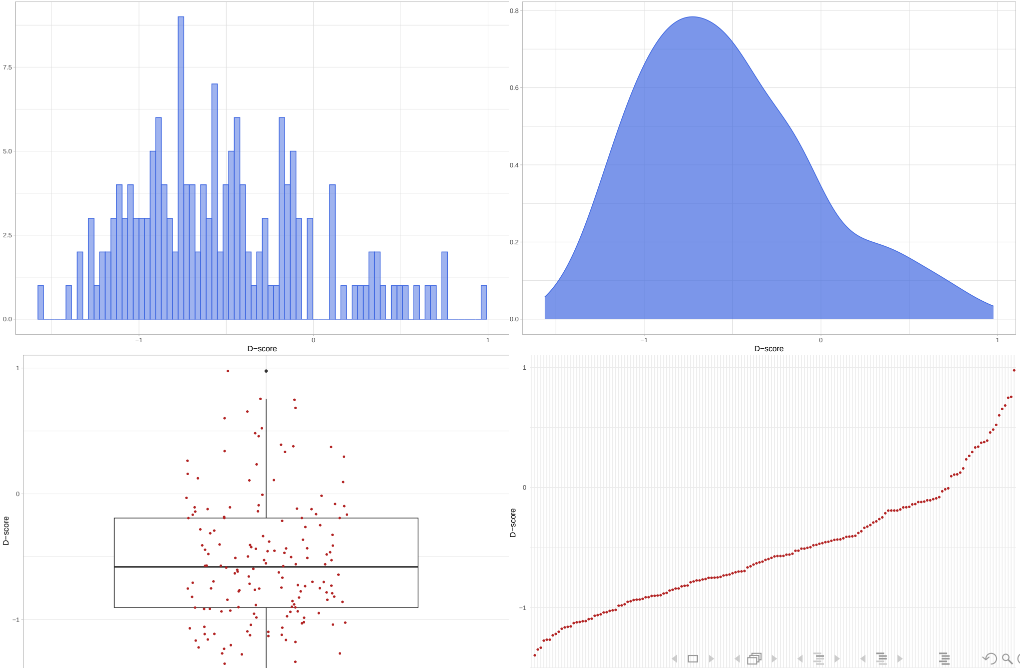
```
$general
       measure value
1 Participants   162
2 Total trials 19440

$statistics
                   measure        mean           sd
1 Response time Mapping A 972.3939630 259.23858664
2 Response time Mapping B 752.5932639 196.28708624
3      Accuracy Mapping A   0.9316592   0.05958240
4      Accuracy Mapping B   0.9655245   0.05919172
5              D-score D1  -0.5271882   0.50846375

$algorithm
[1] "D1"

attr(,"class")
```

```
plot(dscore, graph = "histogram")
plot(dscore, graph = "density")
plot(dscore, graph = "boxplot")
plot(dscore, graph = "points", order_sbj = "D-increasing")
```

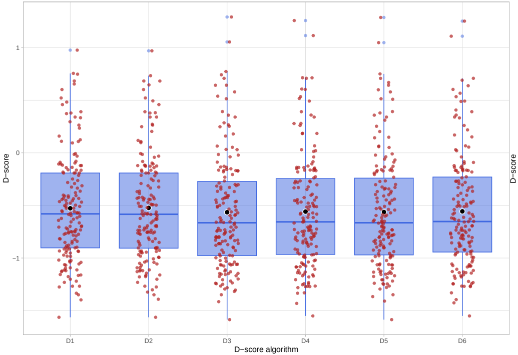


7.5
5.0
2.5
0.0
−1
0
1
D−score
0.8
0.6
0.4
0.2
0.0
−1
0
1
D−score
1
0
−1
D−score
1
0
−1
D−score

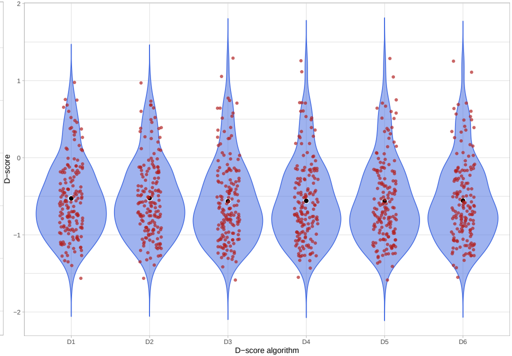
```
multi_iat = multi_dscore(iat_cleandata$data_keep)
str(multi_iat)
```

```
List of 4
 $ scores     :'data.frame':    162 obs. of  7 variables:
  ..$ participant: int [1:162] 4 6 8 11 14 17 18 19 20 21 ...
  ..$ dscore_d1  : num [1:162] -0.762 -1.231 -0.107 -0.953 -0.464 ...
  ..$ dscore_d2  : num [1:162] -0.762 -1.231 -0.107 -0.944 -0.449 ...
  ..$ dscore_d3  : num [1:162] -0.762 -1.25 -0.038 -0.965 -0.593 ...
  ..$ dscore_d4  : num [1:162] -0.7624 -1.2691 0.0202 -0.9721 -0.5739 ...
  ..$ dscore_d5  : num [1:162] -0.762 -1.25 -0.038 -0.955 -0.592 ...
  ..$ dscore_d6  : num [1:162] -0.7624 -1.2691 0.0202 -0.9629 -0.5721 ...
 $ scores_long:'data.frame':    972 obs. of  3 variables:
  ..$ participant: int [1:972] 4 6 8 11 14 17 18 19 20 21 ...
  ..$ score_type : Factor w/ 6 levels "D1","D2","D3",..: 1 1 1 1 1 1 1 1 1 1 ...
  ..$ score      : num [1:972] -0.762 -1.231 -0.107 -0.953 -0.464 ...
 $ source     : chr "IAT"
 $ labels     : chr [1:6] "D1" "D2" "D3" "D4" ...
 - attr(*, "class")= chr [1:2] "multi_dscore" "list"
```

```r
plot(multi_iat)
plot(multi_iat, graph = "violin")
plot(multi_iat, graph = "individual")
```

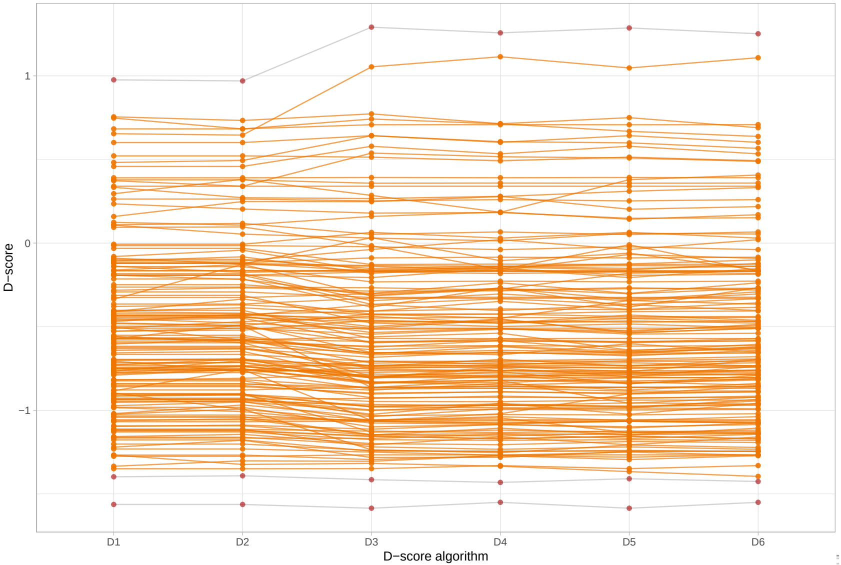


D−score
1
0
−1
D1
D2
D3
D4
D5
D6
D−score algorithm



D−score
2
1
0
−1
−2
D1
D2
D3
D4
D5
D6
D−score algorithm

159 participants involved in at least one rank inversion



Excercise

Prendete i dati disponibili sul sito del corso e importateli in R:

```
dati =  read.delim("data/iat_data.dat")
str(dati)
```

```
'data.frame':   16128 obs. of  7 variables:
 $ id      : int  1 1 1 1 1 1 1 1 1 1 ...
 $ order   : int  1 1 1 1 1 1 1 1 1 1 ...
 $ test    : int  1 1 1 1 1 1 1 1 1 1 ...
 $ blocknum: int  3 3 3 3 3 3 3 3 3 3 ...
 $ trialnum: int  1 2 3 4 5 6 7 8 9 10 ...
 $ correct : int  1 1 1 1 1 1 1 1 1 1 ...
 $ latency : int  812 687 723 619 657 580 596 687 618 936 ...
```

Sono dati di uno IAT usato nel paper

Röhner, J., & Thoss, P. J. (2019). A tutorial on how to compute traditional IAT effects with R. *The Quantitative Methods for Psychology, 15*(2), 134–147.

e sono accessibili su OSF

Non si hanno i nomi dei blocchi, ma solo

```r
table(dati$blocknum)
```

```
   3    4    6    7
4032 4032 4032 4032
```

I blocchi 3 e 4 saranno la condizione A e i blocchi 6 e 7 la condizione B

Calcolare i punteggi IAT con il pacchetto `implicitMeasures` e investigare le possibili differenze attribuibili alla variabile `order`